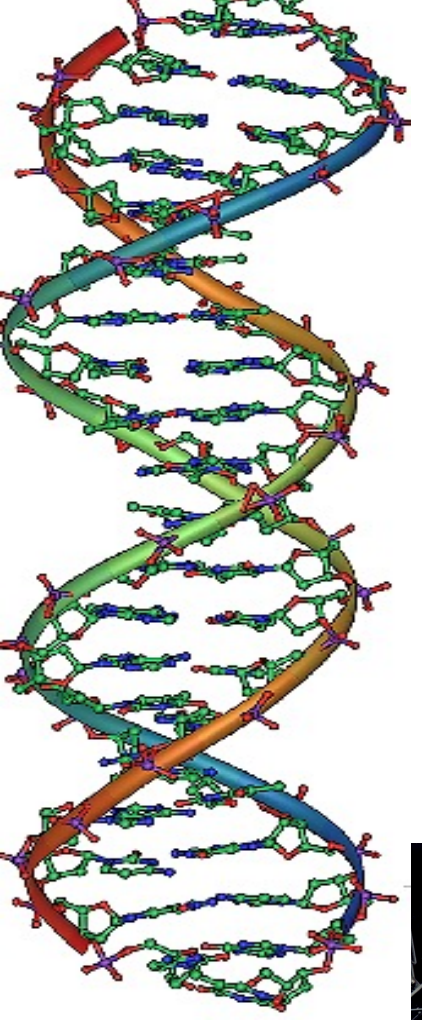
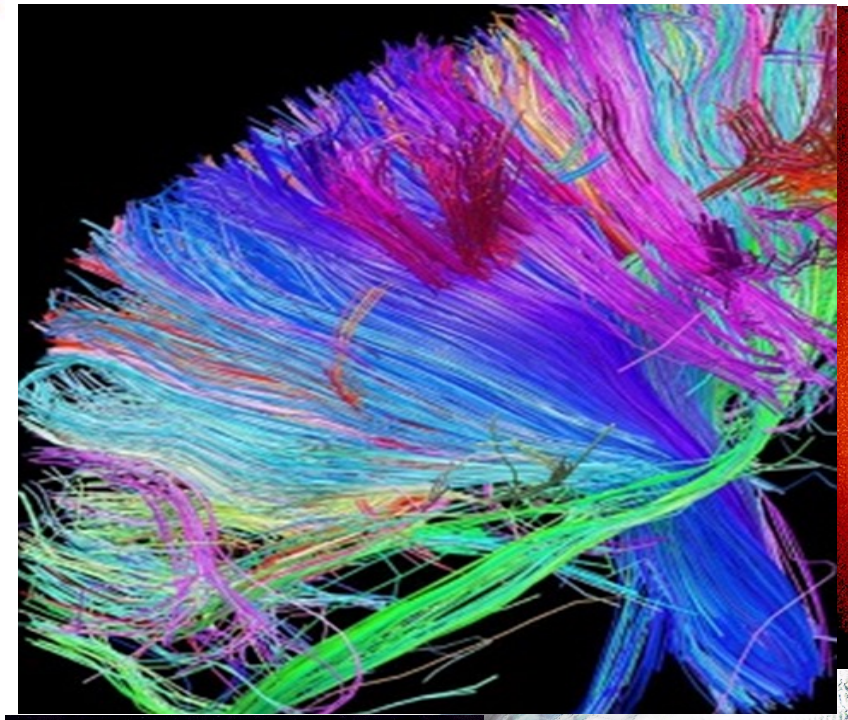
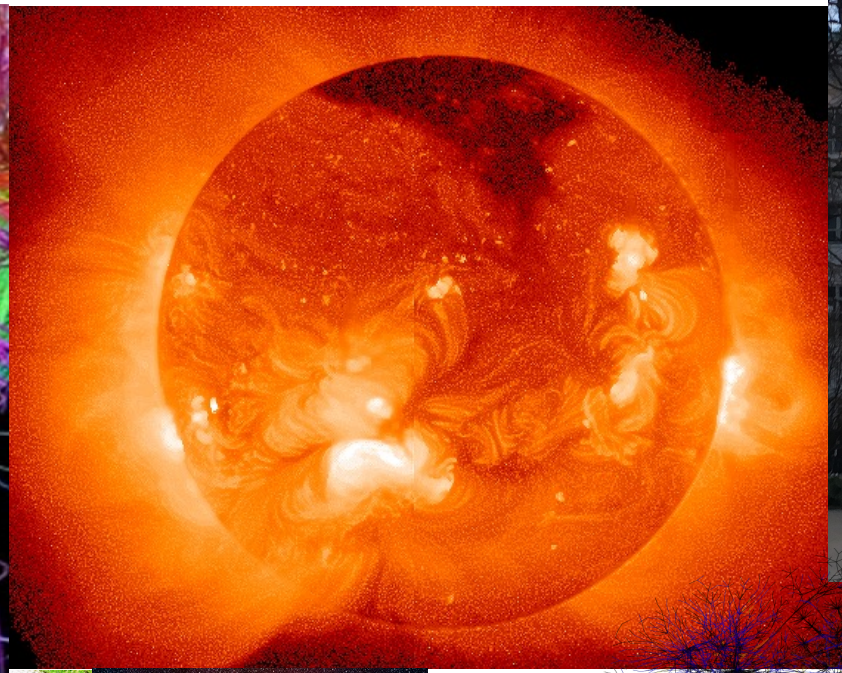
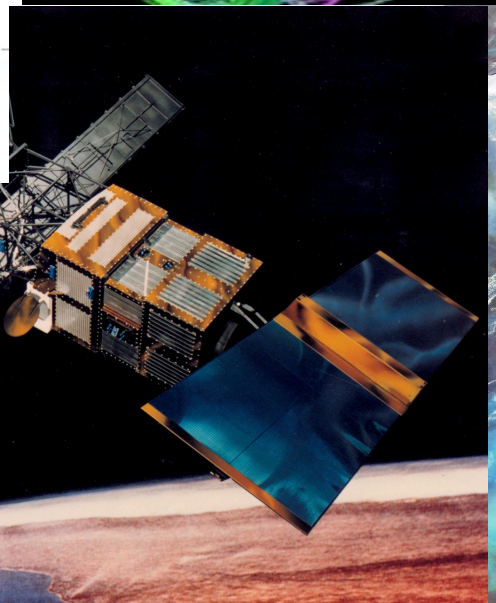
Ch. 5.3: The Innovation Policies.

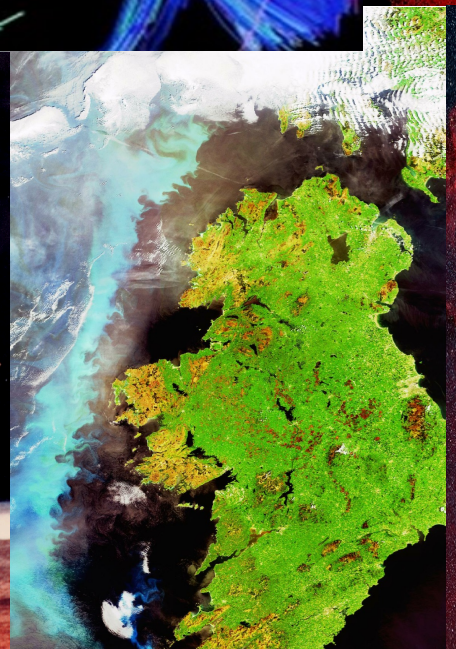


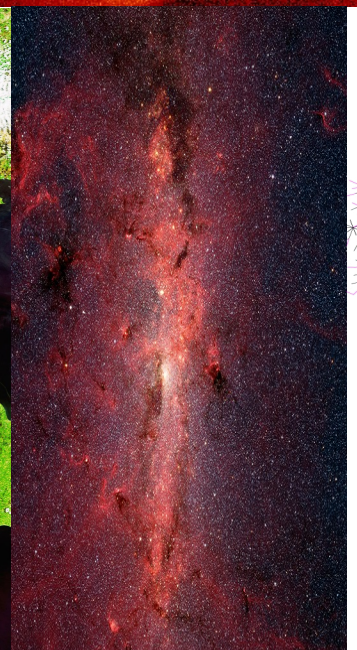


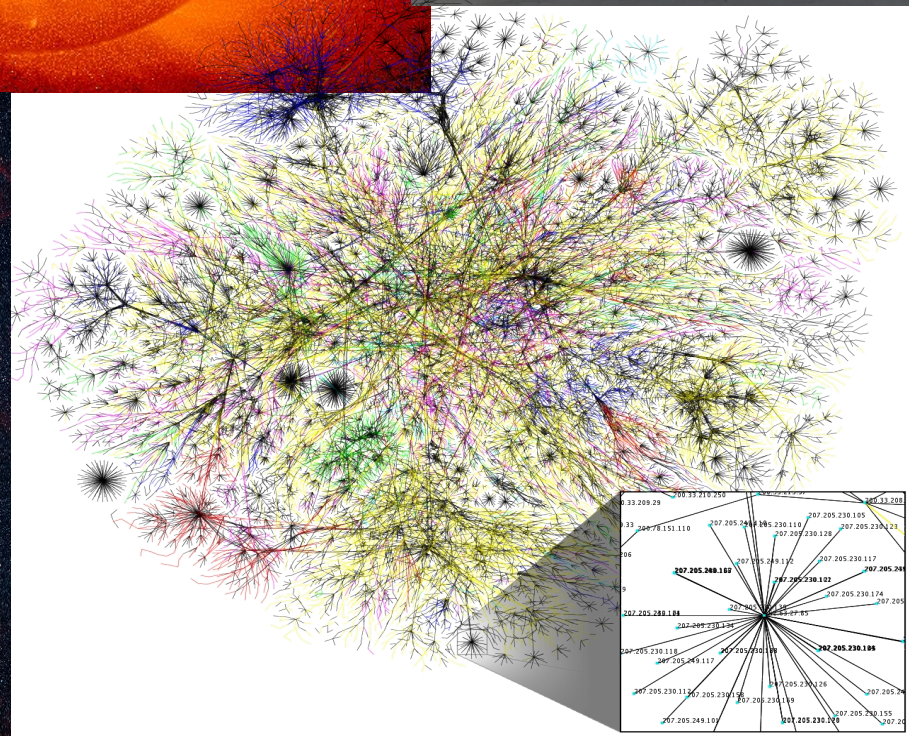






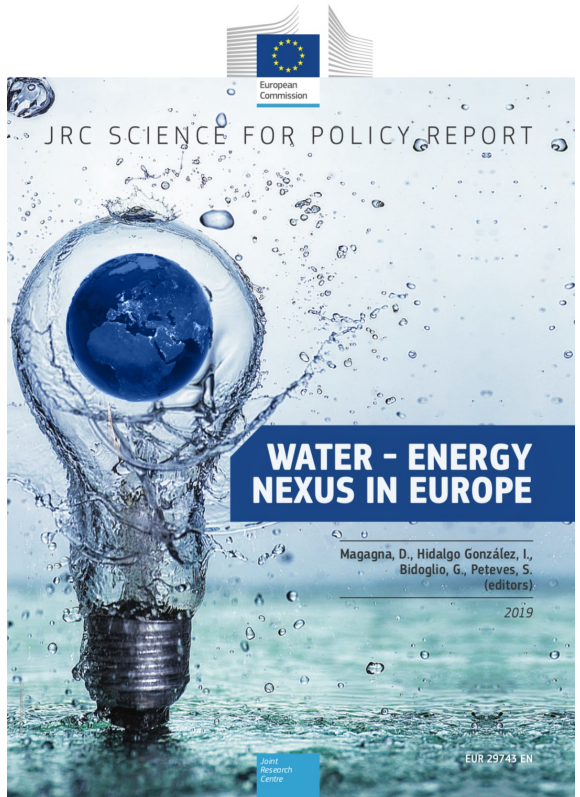






Innovation Policies – Research.

- **European Community direct research activities:**
- **1958 Euratom** — the European Atomic Energy Community – a joint research program to exploit nuclear energy for peaceful purposes.
- As part of this, the Joint Research Centre (JRC) was created: The „EU Science Hub“.
- The JRC has now 7 in institutes in IT, BE, NL, DE and ES and 3.000 staff.
- Its task is to provide scientific support to the EU policy makers in all major policy areas: environment, consumer protection, trade, energy, economics……
- Most activities performed in cooperation with the best experts in the MS.
- **Examples for reports/activities:**
  - Short and longer-term effects of European regional policy.
  - Global value chain analysis of the EU automotive sector.
  - EU Green House Gas Inventory (jointly with EEA).
  - European Flood Alert System.
  - European Environmental Information System (with EEA, EUROSTAT).
  - Crop monitoring in Europe and neighbouring countries.



Innovation Policies – Research.

- **Funding of joint research between Member State institutions:**
- **Horizon Europe** (2021- 2027) - 100 billion EUR, 3 pillars:

– **Open Science:** European Research Council (ERC) grants for top-level fundamental research, Marie-Sklodowska-Curie-Program for mobility of researchers (post-doc), support of research infrastructure.

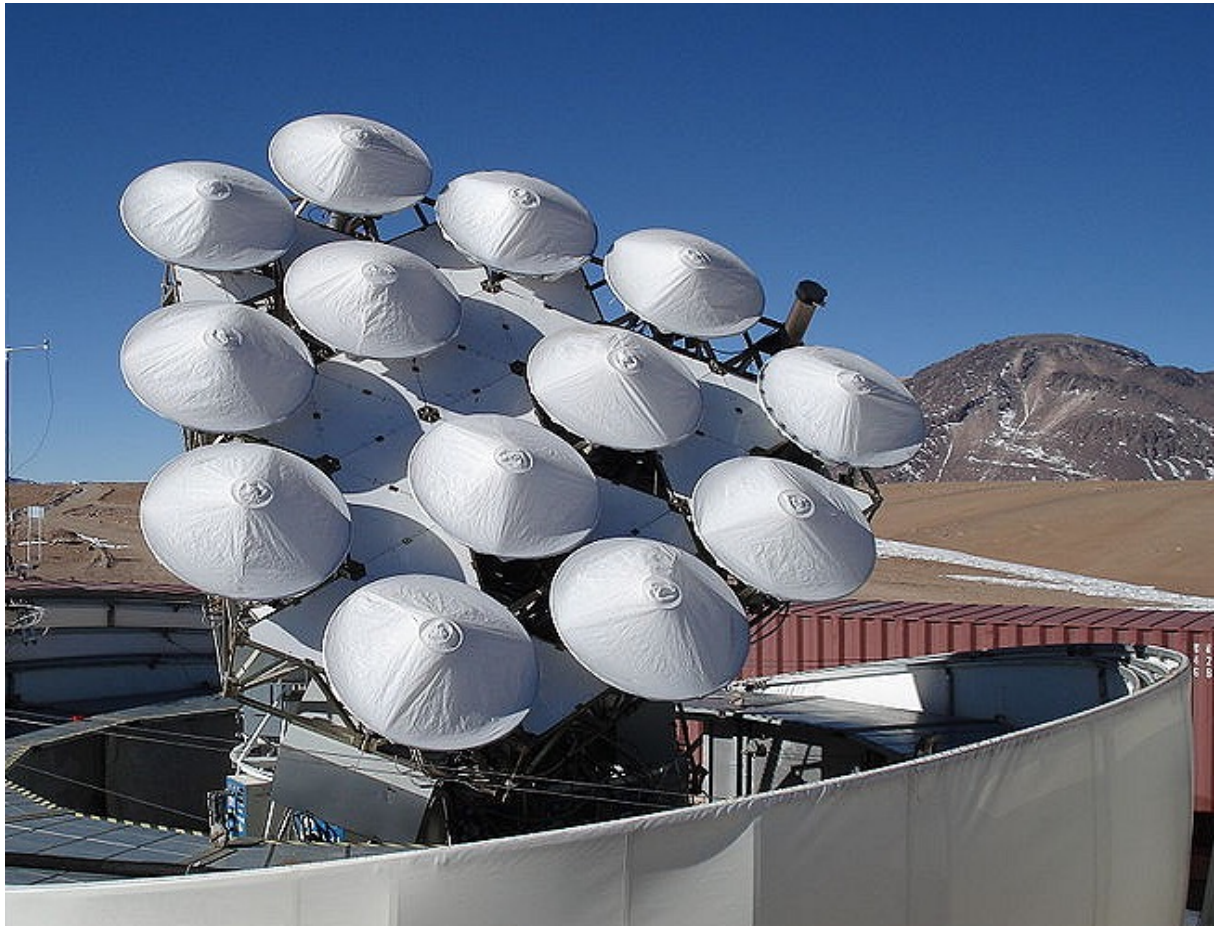
– **Targeted research:** carried out in multinational consortia with typically 6 – 10 institutions cooperating to enhance global competitiveness of European industry and to meet global challenges: focus on digitalisation, space industry, climate, energy and mobility, natural resources, food, health.

– **Open innovation:** European Innovation Council (EIC), European Institute for Innovation and Technology – Cooperation of leading Technical Universities in the EU.

The EU funding of research in Europe amounts to only 5 % of the total spending for research and innovation in the Member States.

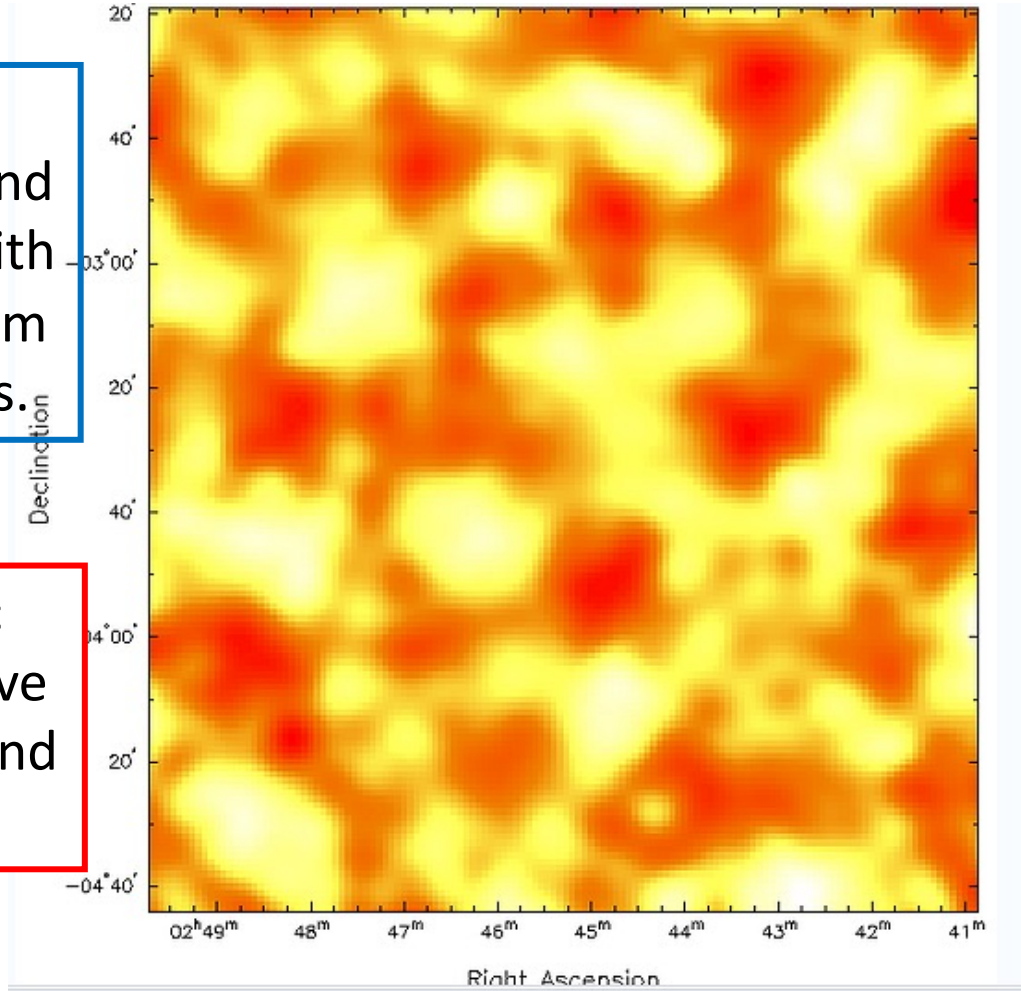
Large Scale Research Facilites:
European Southern Observatory (ESO).

- Very Large Telescope (VLT) in Atacama Desert in Chile :
  - 12 antennas, diameter each 1,4 m for measuring the relict cosmic microwave radiation from the „Big Bang“ occurring 13,73 billion years ago giving birth to our universe.



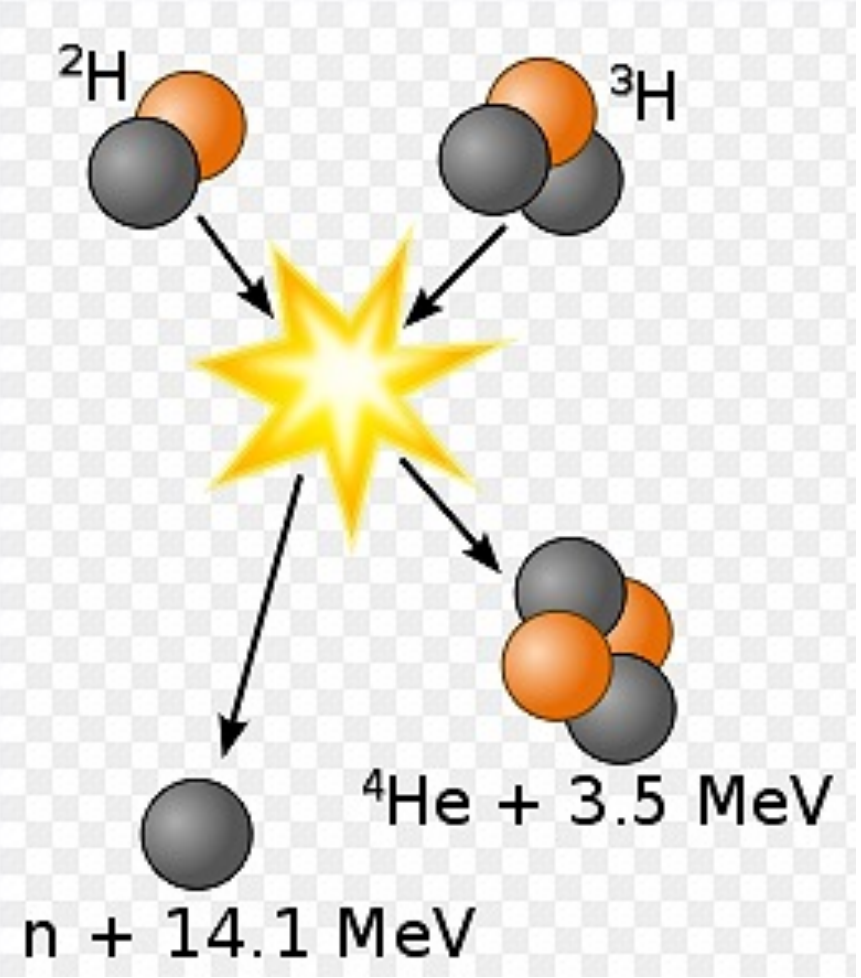
Cosmic Background Imager with twelf 1.4 m antennas.

Cosmic Microwave Background Image.

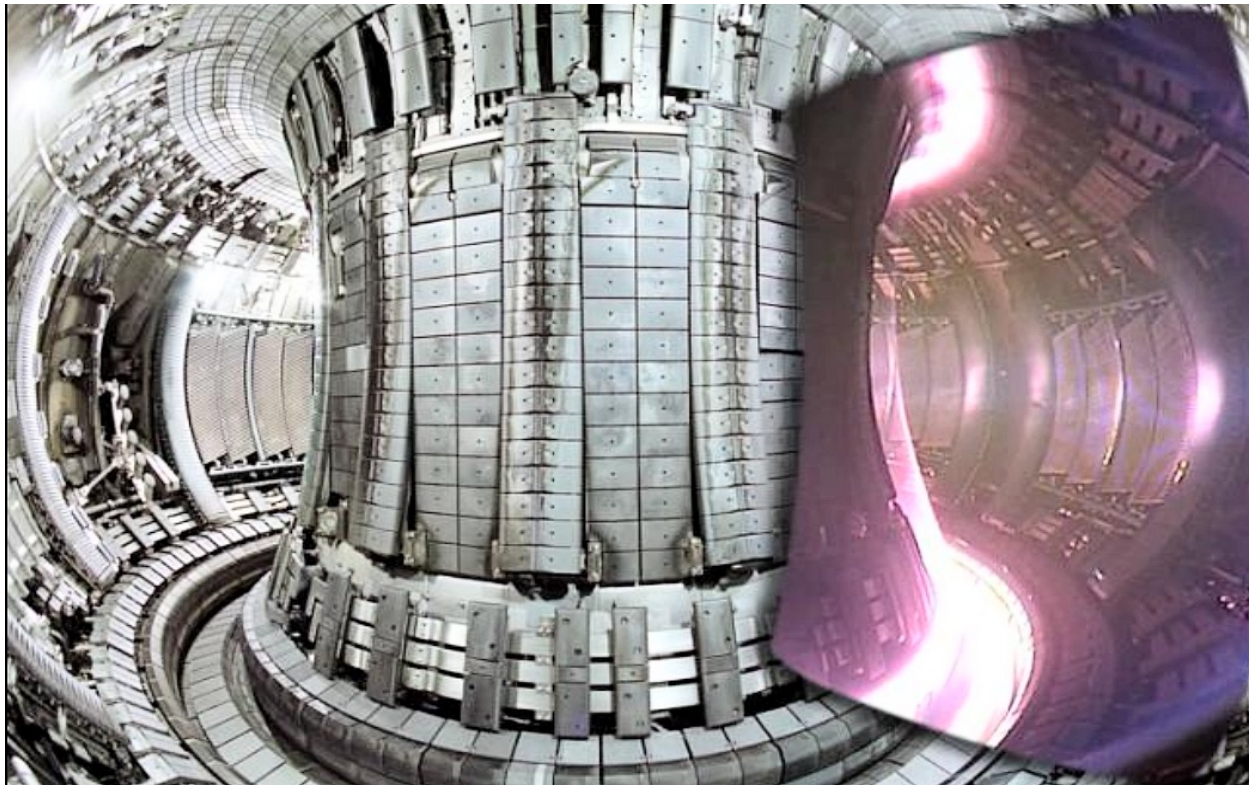


Large Scale Research Facilites:
The International Thermonuclear Experimental Reactor (ITER).

- **Development of a fusion reactor for electricity production:**
- Fusion is the process at the core of our Sun.
- Hydrogen nuclei collide, *fuse* into heavier Helium atoms and release tremendous amounts of energy in the process.
- The most efficient fusion reaction is between the two Hydrogen isotopes Deuterium (D) and Tritium (T), which however requires temperatures of 150 million °C to take place.
  - When deuterium and tritium fuse, two nuclei come together to form a helium nucleus (an alpha particle), and a high-energy neutron.
  - The reaction occurs only if the 2 nuclei get within 100 femtometer (1×10^{-13} meter) of each other.
- In ITER, this distance of approach is made possible by high temperatures and magnetic confinement.



- At extreme temperatures, electrons are separated from nuclei and a gas becomes a **plasma** - a hot, electrically charged gas.
- The plasma is confined by means of magnetic fields in a vacuum vessel (a double walled steel container called tokamak).
- The neutrons generated in the fusion reaction transfer their energy to the walls as heat, which will be dispersed through cooling towers to produce steam and - by way of turbines- electricity.

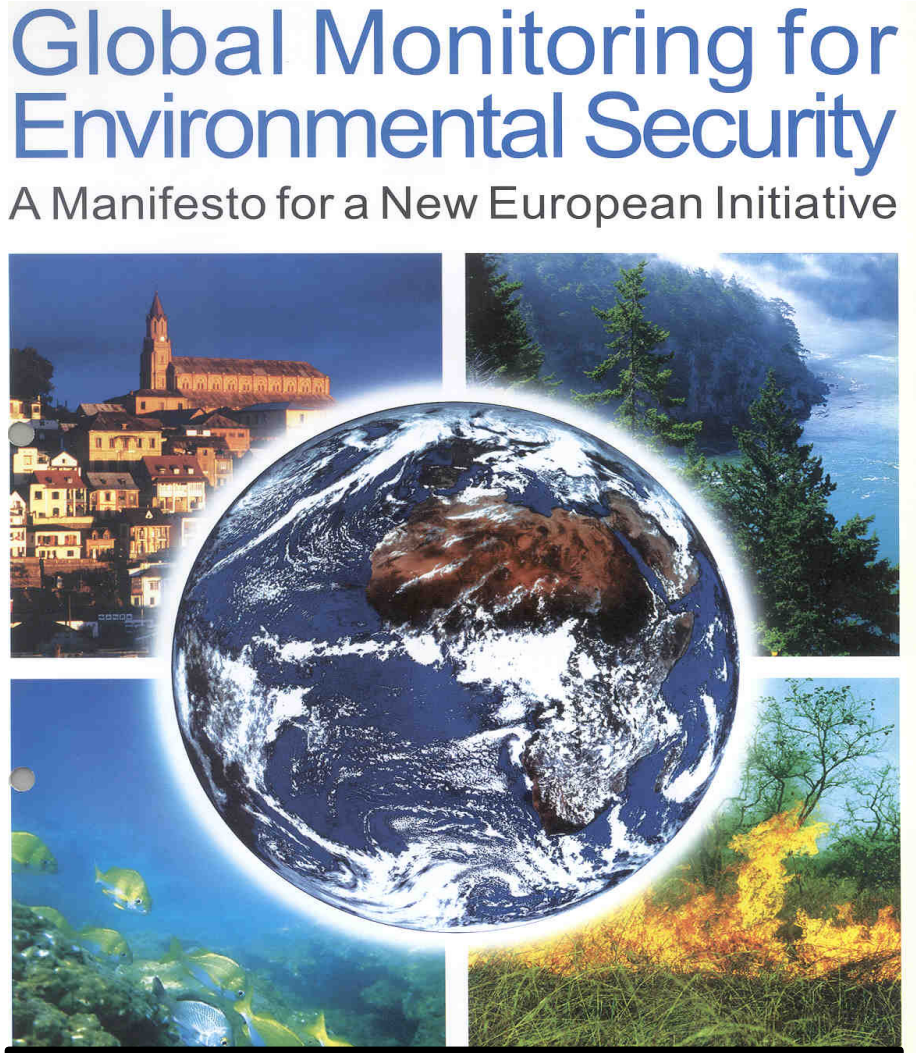


- **Fundamental problems:**
  - Walls of reactor have a very limited lifetime (7 years ?) due to the heavy bombardment with neutrons.
  - Production of radioactive tritium during process.
- **Site and manpower:** Cadarache, France; 3.000 staff
- **Timeline:** construction finished in 2034, operation for 30 years ?
- **Costs:** 25 billion EUR ?

Planned ITER tokamak: external diameter 20 m, internal diameter 6,5 m.

Innovation Policies:

Copernicus (GMES).



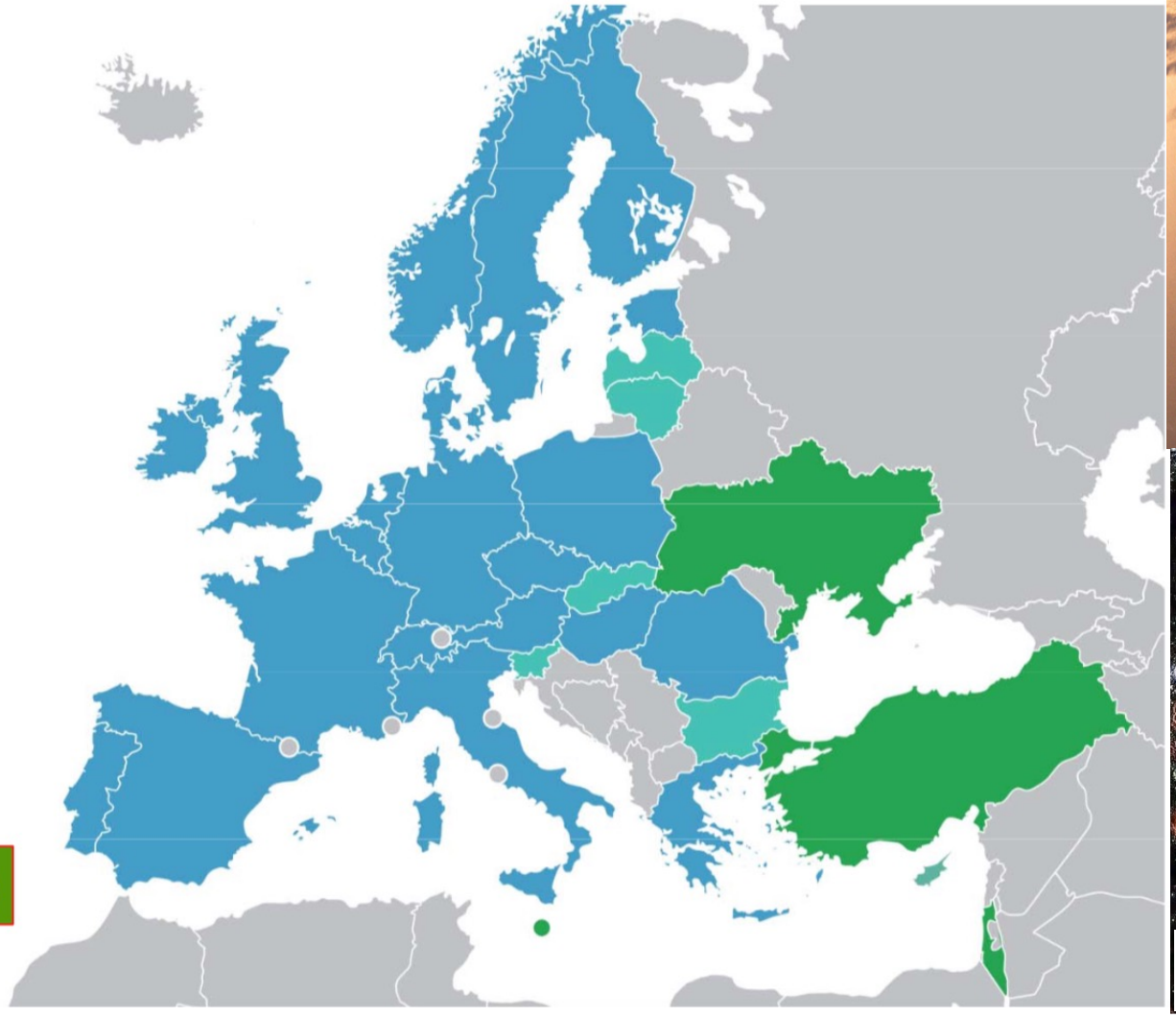
The Baveno Manifesto on GMES initiated by the JRC 1996.

- Joint undertaking of EU and ESA (European Space Agency) to develop and operate an independent European capacity for earth observation to support EU and national policies.
- Copernicus consists of 30 satellites circulating the earth at a height of 700-800 km with a width of the observation path of 300-1.000km.
- Each satellite is equipped with a sophisticated sensor system operating in a spectral range from microwave to UV radiation delivering important information for monitoring land, sea and atmosphere, climate variables and crisis situations.
- GMES is the European contribution to a comprehensive international earth observation system comprising eventually all major satellite operating nations like USA, Japan, China, India, Russia (GEOSS).

European Space Agency.

Intergovernmental organisation of 22 States dedicated to the explaration of space.

- Founded 1975
- Headquarter: Paris
- Sites: Nordwijkershout (NL), Darmstadt (DE), Frascati (IT), Harwell (UK), Kourou (French Guiana)
- Budge 8 billion EUR annually

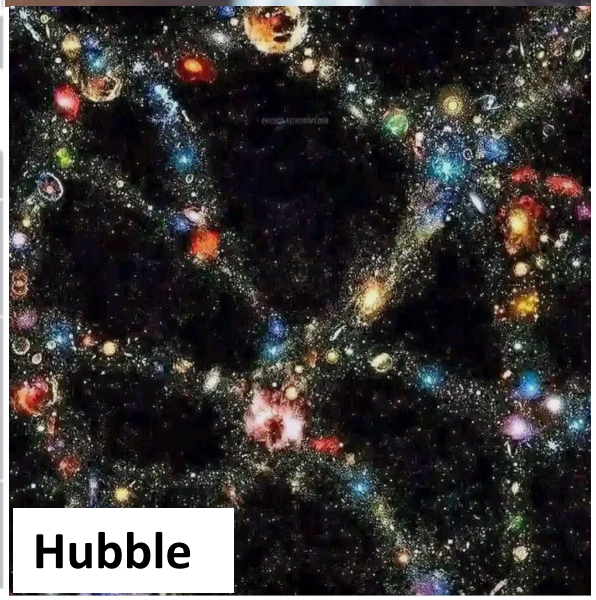


Member States

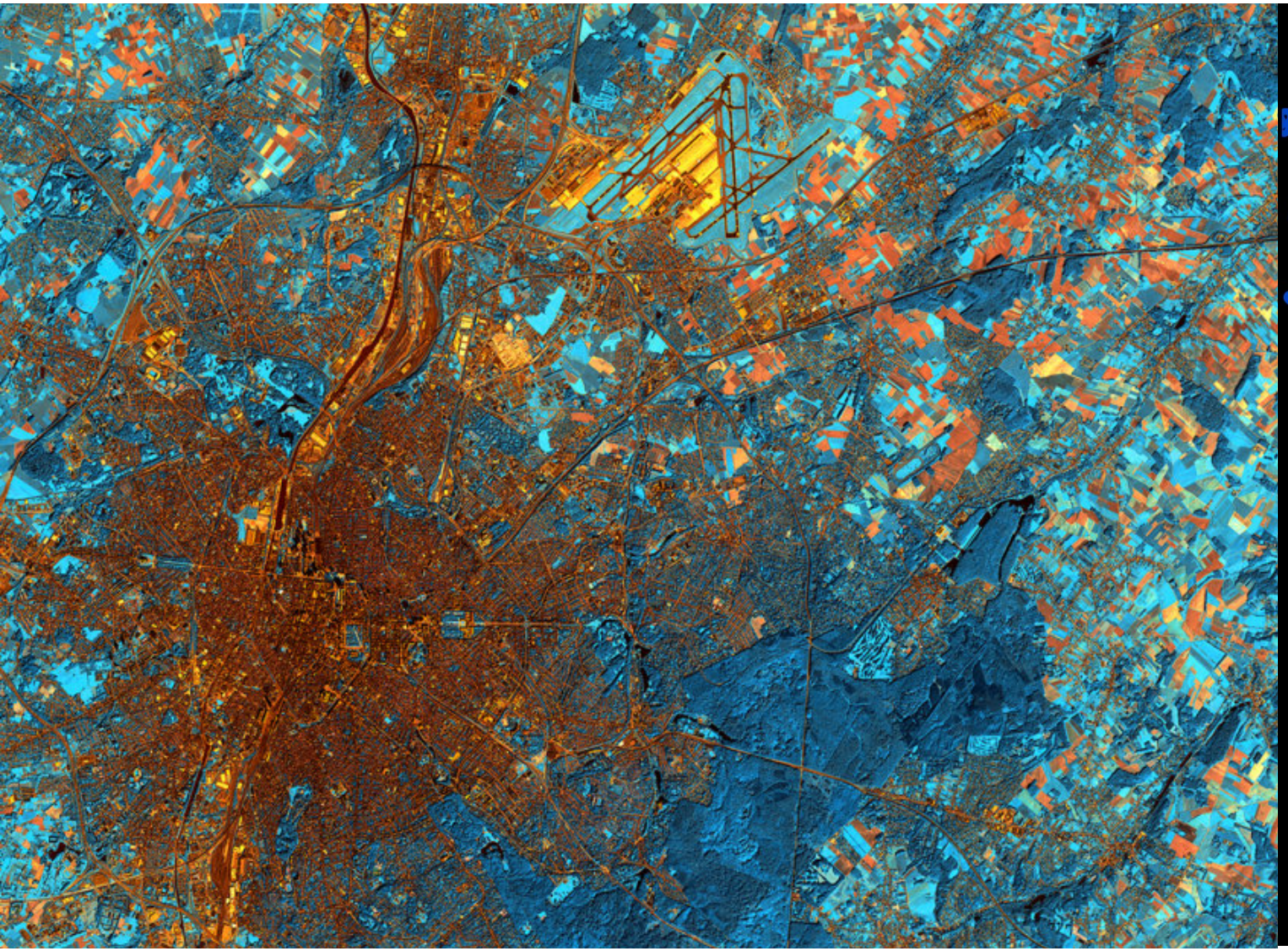
Cooperation Agreements

Source: ESA

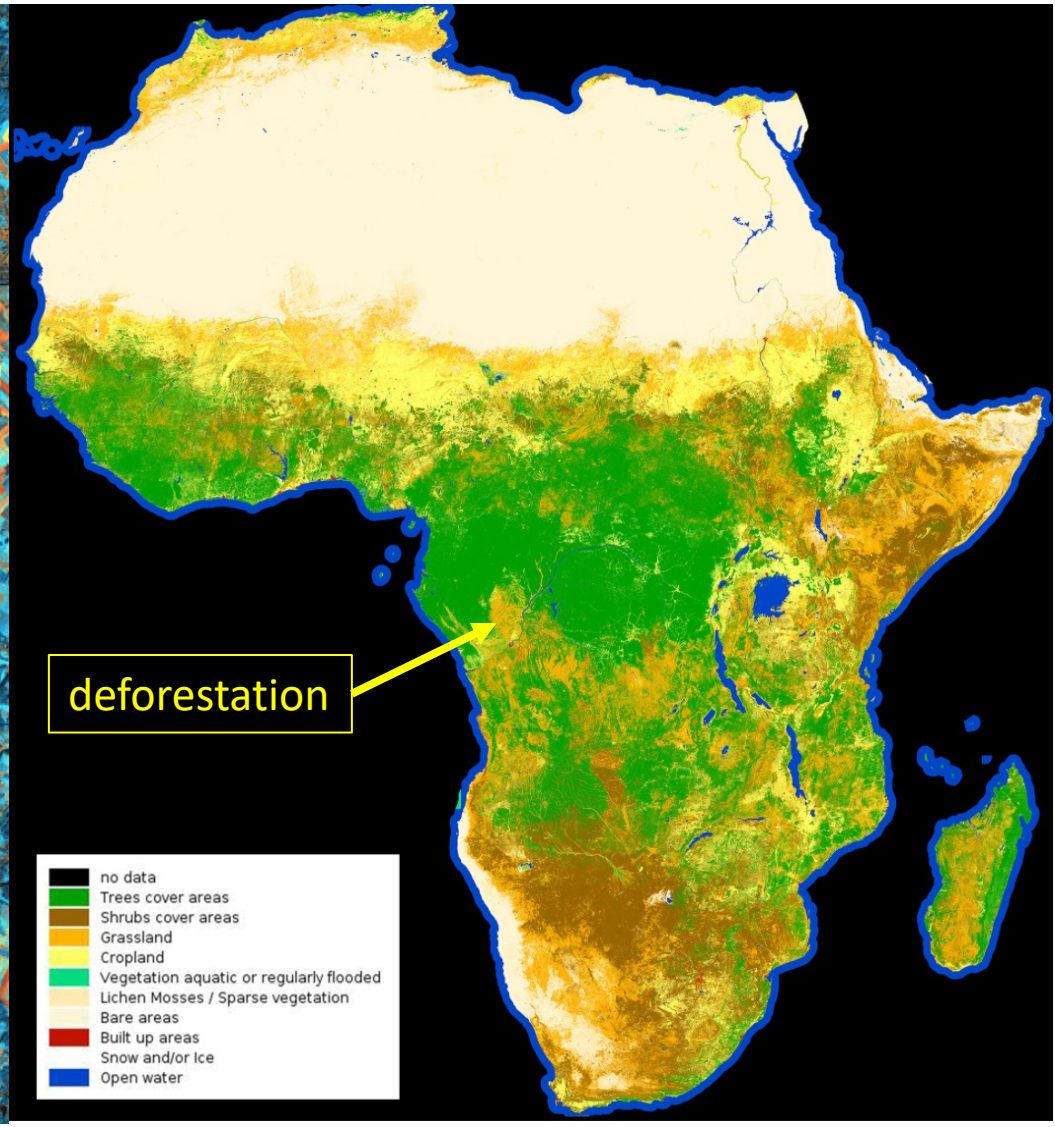




Copernicus Services:
Land Monitoring.

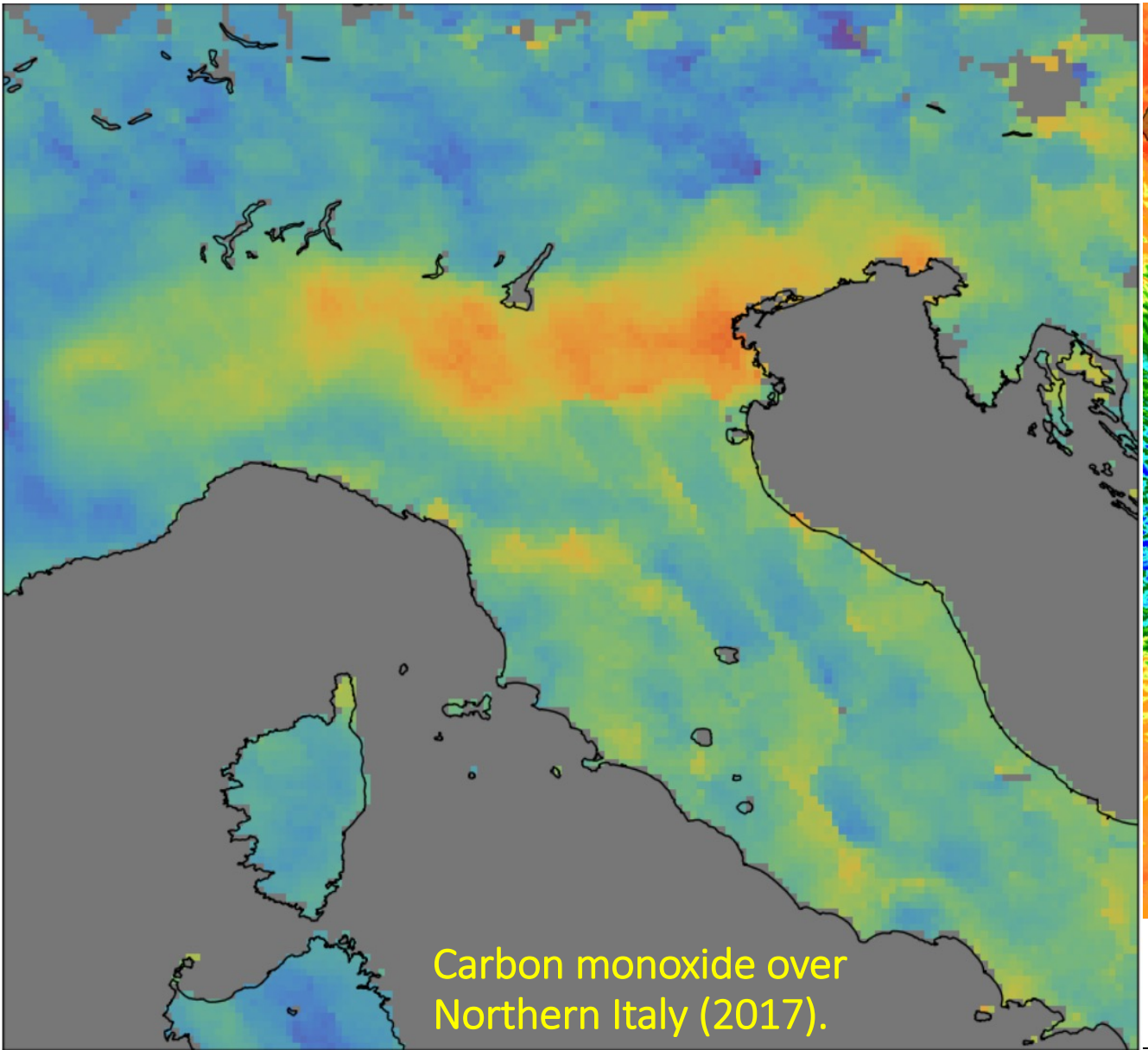


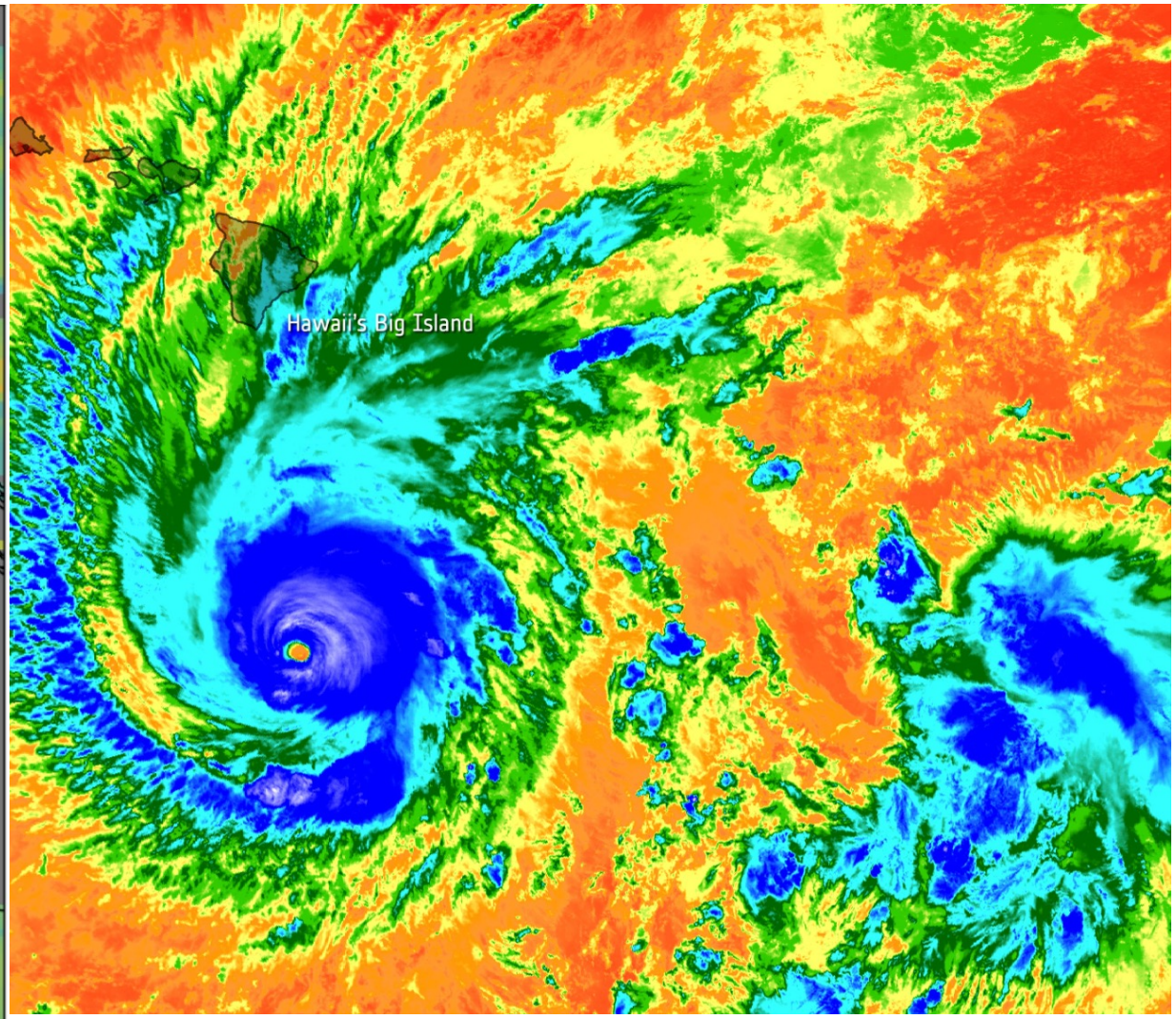
BRUSSELS false-colour image (resolution 10 m).



Land cover Africa – true color image.

Copernicus Services:
Atmospheric Monitoring.

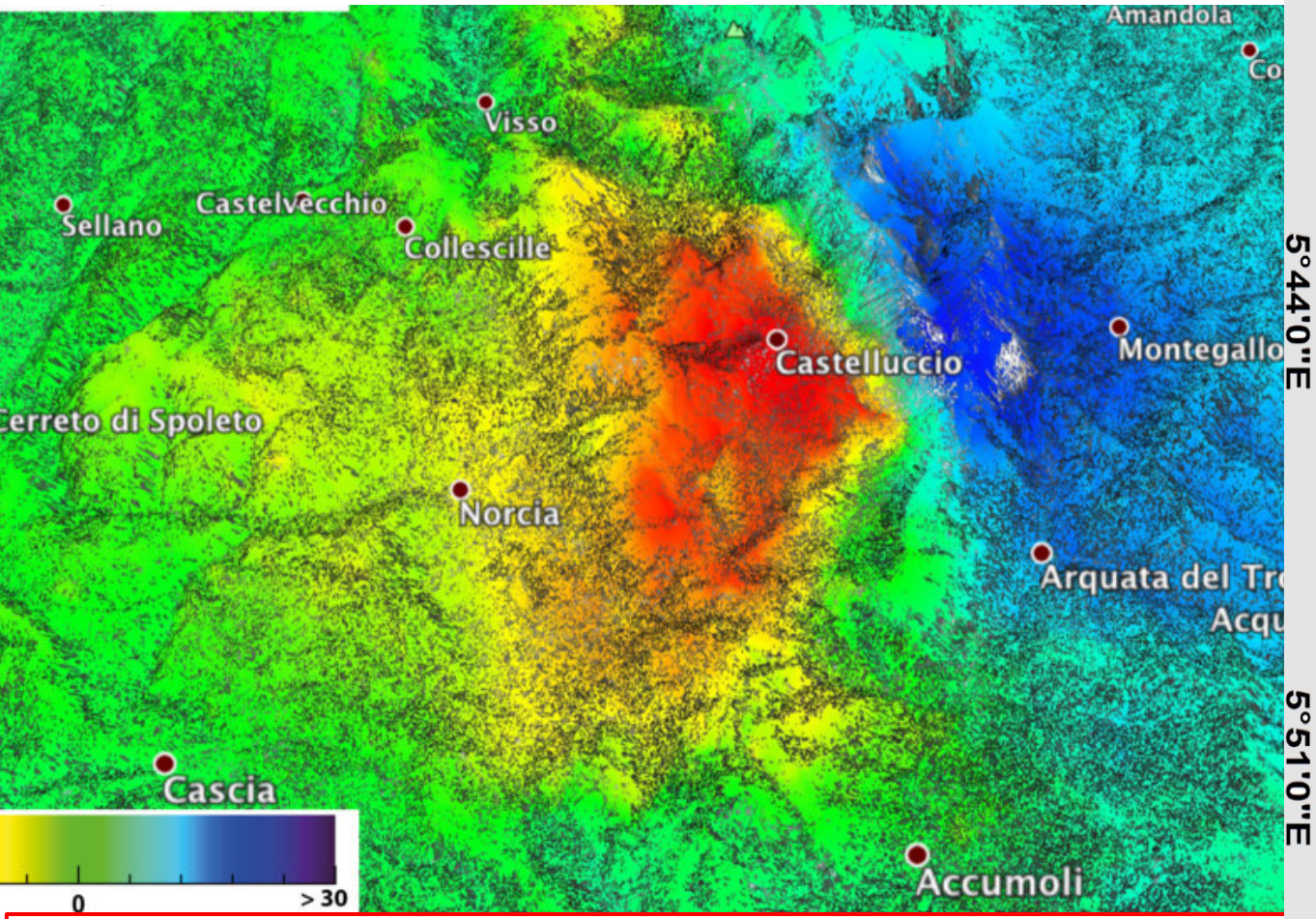




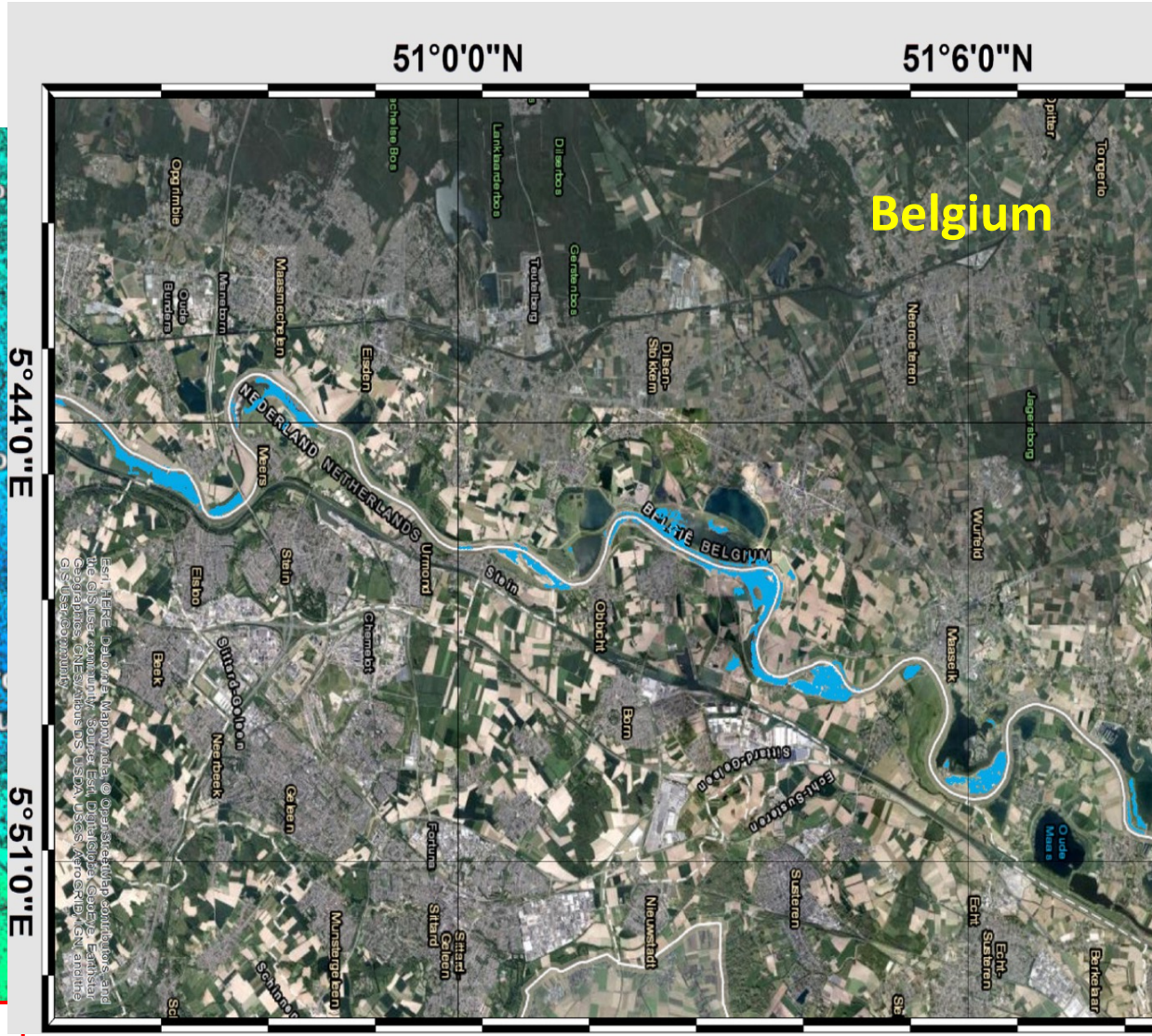
Hurricane Lane near Hawai Aug. 2018
Blue: - 80°C, orange + 15°C.

Copernicus Services:
Crisis Management.

EARTHQUAKE in ITALY 30 OCTOBER 2016 (magnitude 6,5)



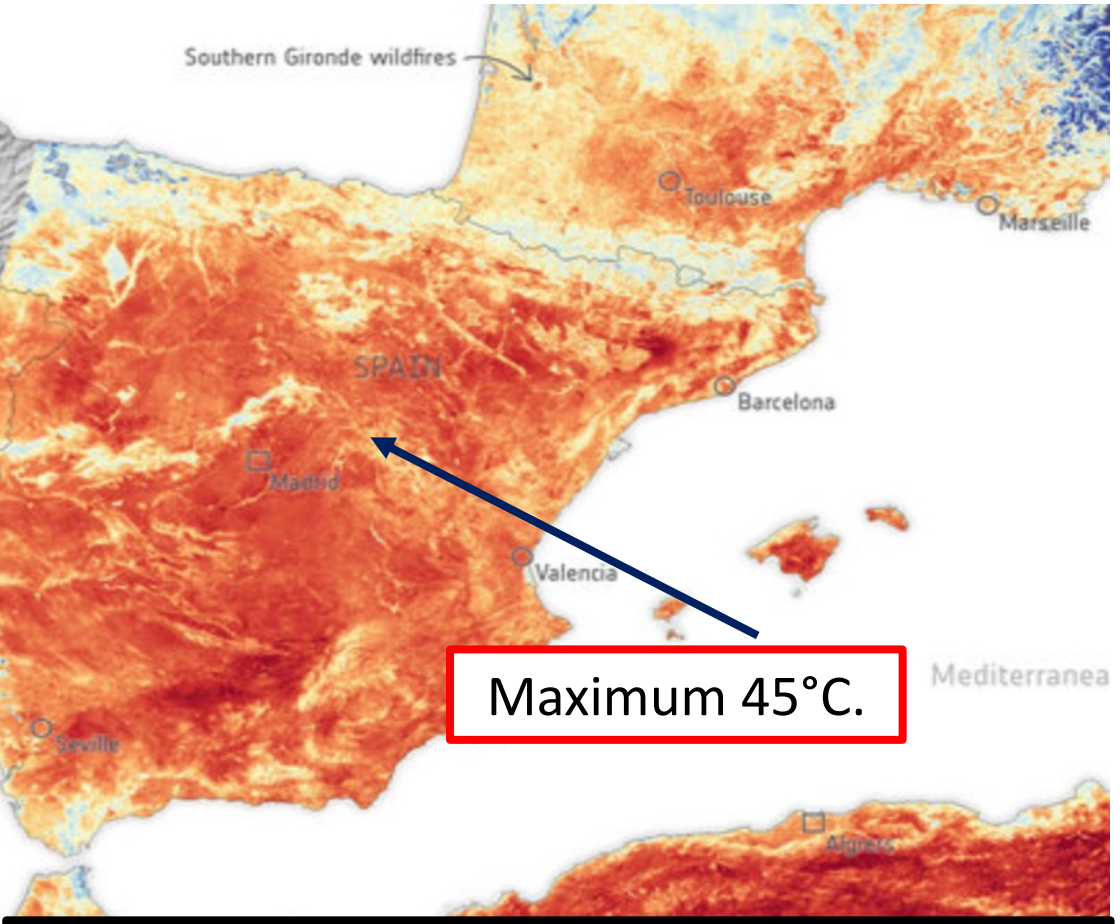
Ground deformations extending across 130 km^2 with a maximum vertical displacement of about 70 cm.



Mapping of floods caused by the river Maas 2021.

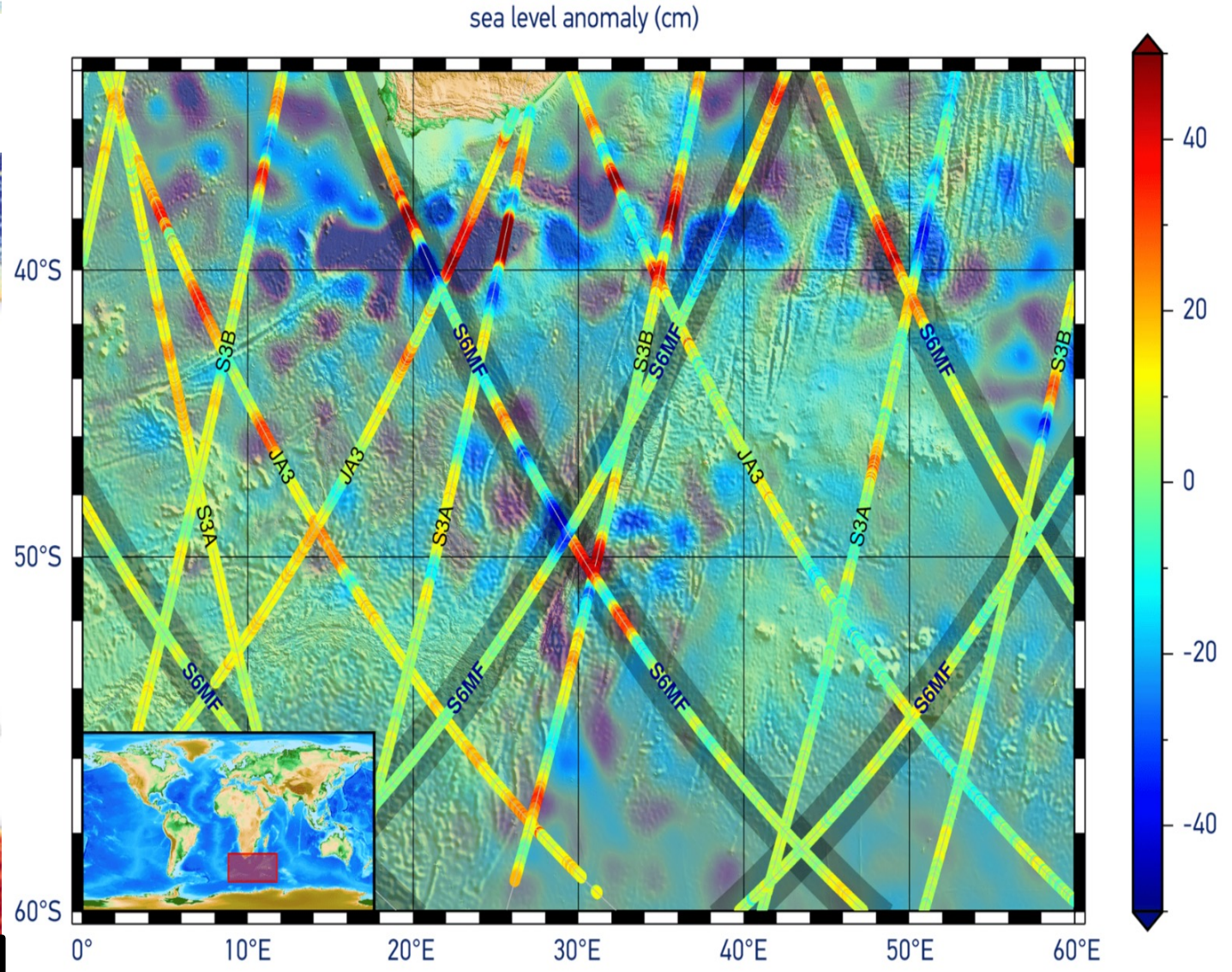
Copernicus Services:
Climate Change Effects.

Measurement of land surface temperatures
20 July, 2022 with an accuracy of 0,3°C.



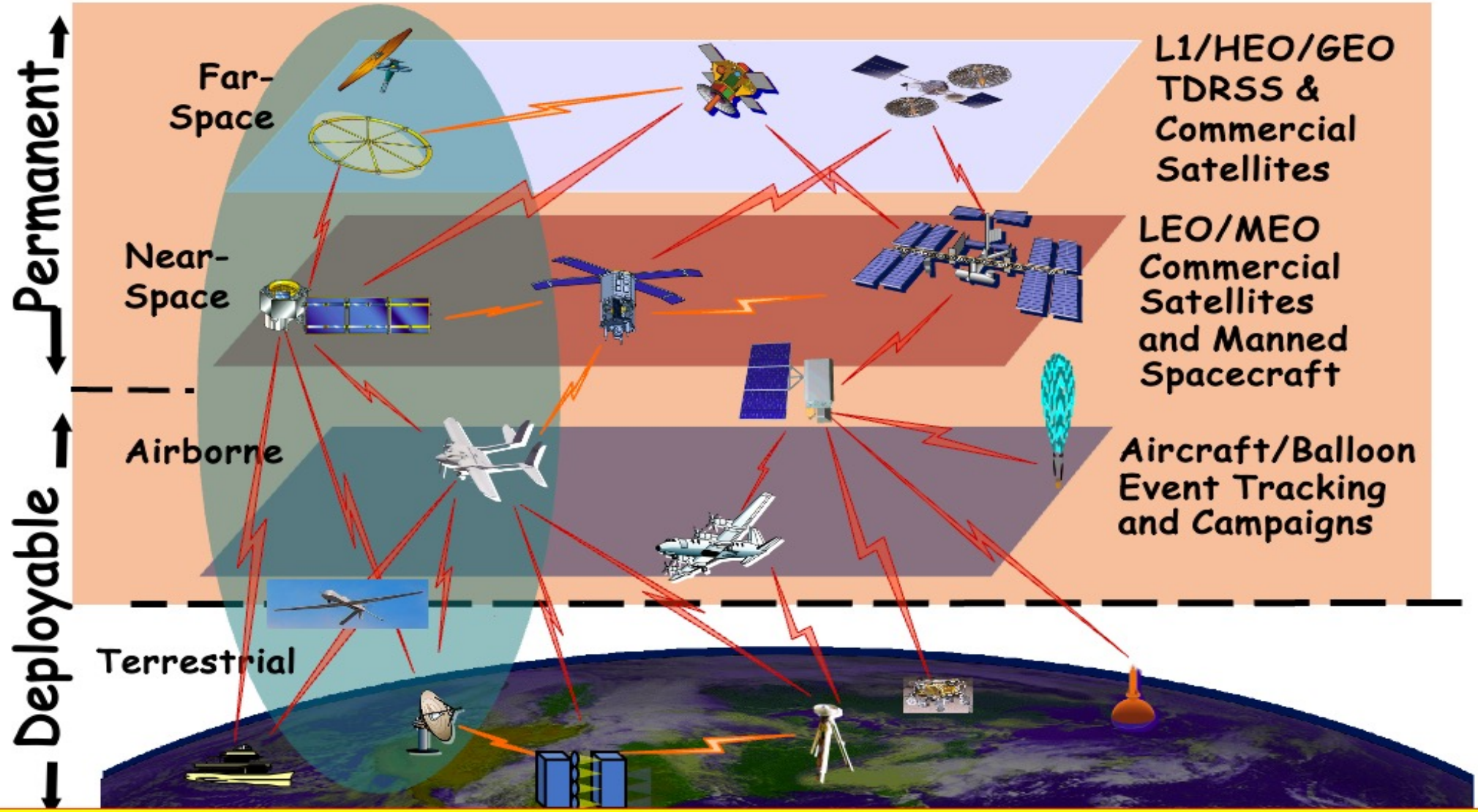
Maximum 45°C.

Attention: land surface temperatures are much higher than the metrological air temperatures.

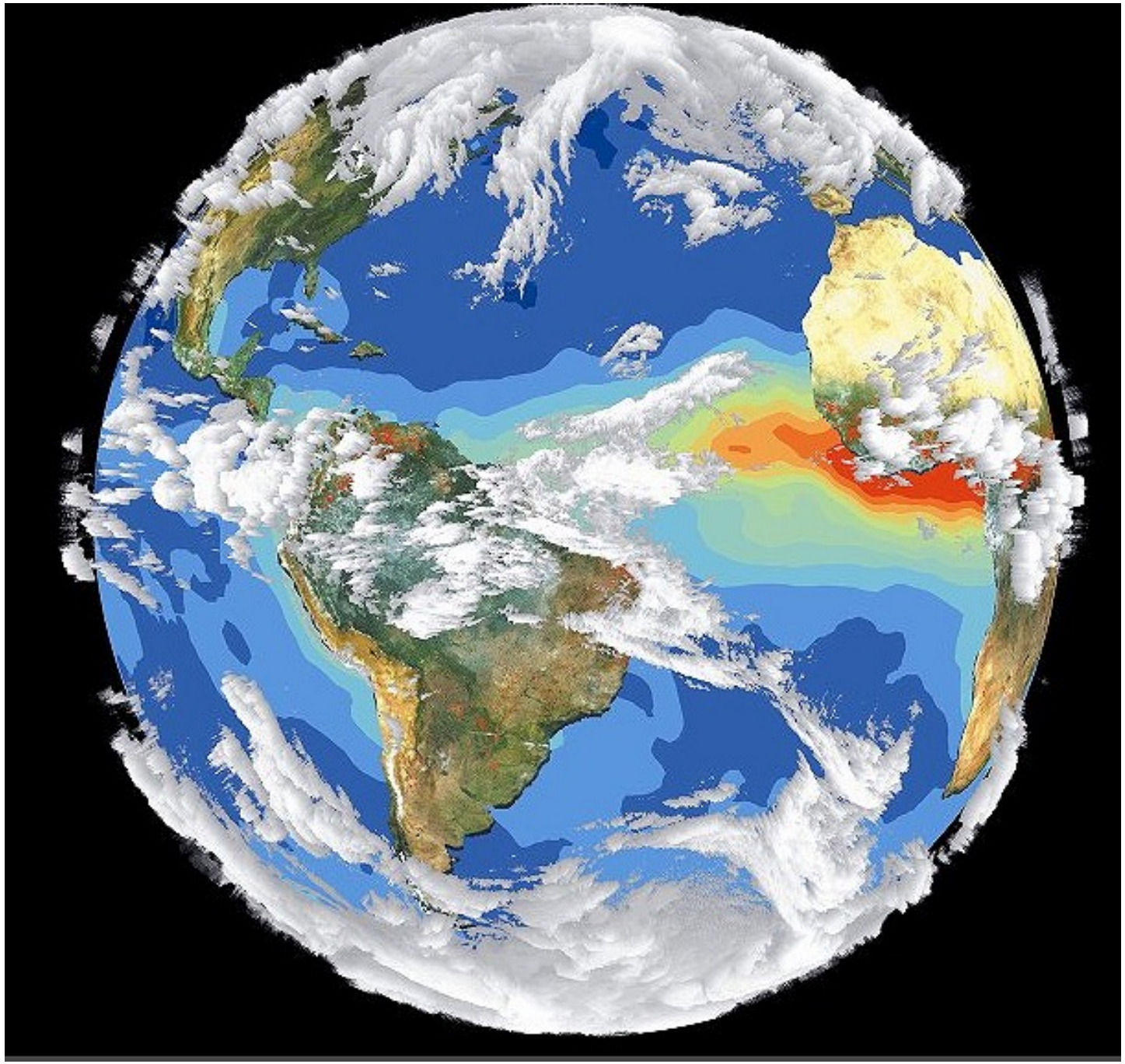


Radar-Altimetry: Precision measurements of sea level rise.

Global Earth Observation: GEOSS.



GEOSS: Global Earth Observation System of Systems
(USA, EU, Japan, China, India, Brazil, South Africa, Korea……). Source: NASA



Global Earth Observation: GEOSS.

- Composite image taken from 6 different satellites:
  - Land image in true color from SeaWiFS of NASA.
  - Oceanic aerosol caused by biomass burning and windblown dust over Africa by NOAA.
  - Clouds from infrared images from four geostationary weather satellites (NOAA's GOES 8 and 9, the ESA's METEOSAT, and Japan's GMS 5).

Innovation Policies:
Education (Brain Circulation).

- ERASMUS is a university exchange programme which began in 1987.
- 4.000 higher institutions from 37 countries participating.
- Every year more than a million students receive an ERASMUS grant (amounts to roughly 5% of the European graduates).
- 16 million students have received an ERASMUS grant and spent a part of their course studying in another country.
- ERASMUS course credits are fully recognised by the students' own institution.
- ERASMUS has also been responsible for changing the direction of European higher education policy ('Bologna Process' – established Bachelor as first and Master as second degree).
- Financial endowment of ERASMUS for period 2021-2027: 30 bio EUR



Desiderius Erasmus
Roterodamus
1466 - 1536

European Deficits in Innovation.

- **The European Union is lagging behind the USA – and in the meantime also China – in important areas:**
  - Microelectronics (chips, devices)
  - Software engineering
  - Artificial intelligence
  - Electromobility (batteries)
  - Genetic engineering
  - Cyber defense
  - Military technology
  - Communication means (What's App….)
  - Social media with revolutionary impact on governance (Facebook, Twitter/X, TikTok….)

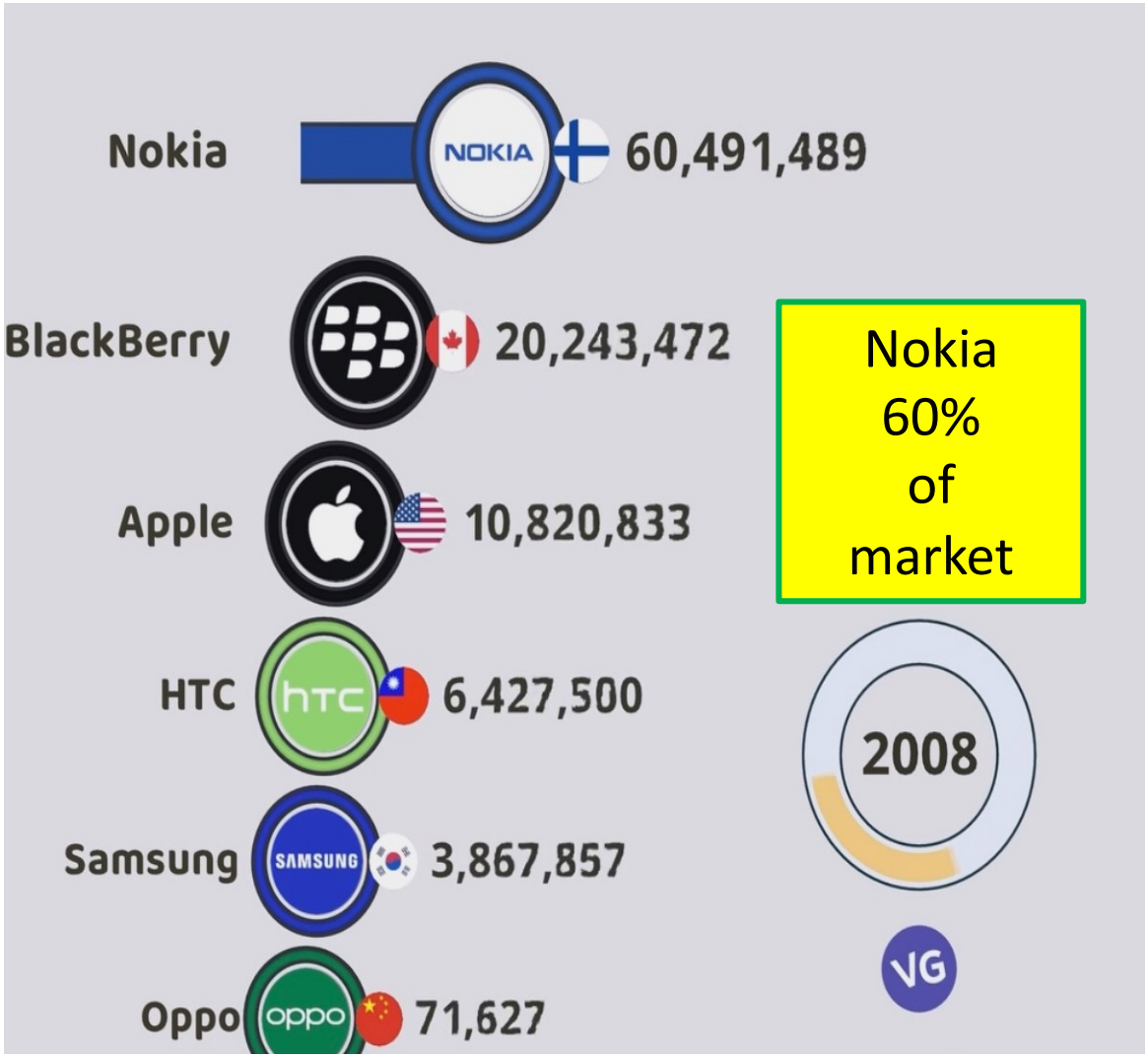
- **Problems:**
  - Insufficient financing of research (2,5% of GDP, USA 4,5%, China 3,5%,
  - slow transformation of research outcomes into innovative products,
  - lower profit margins of companies,
  - lack of venture capital,
  - excessive bureaucracy,
  - pessimistic approach towards science and technology……

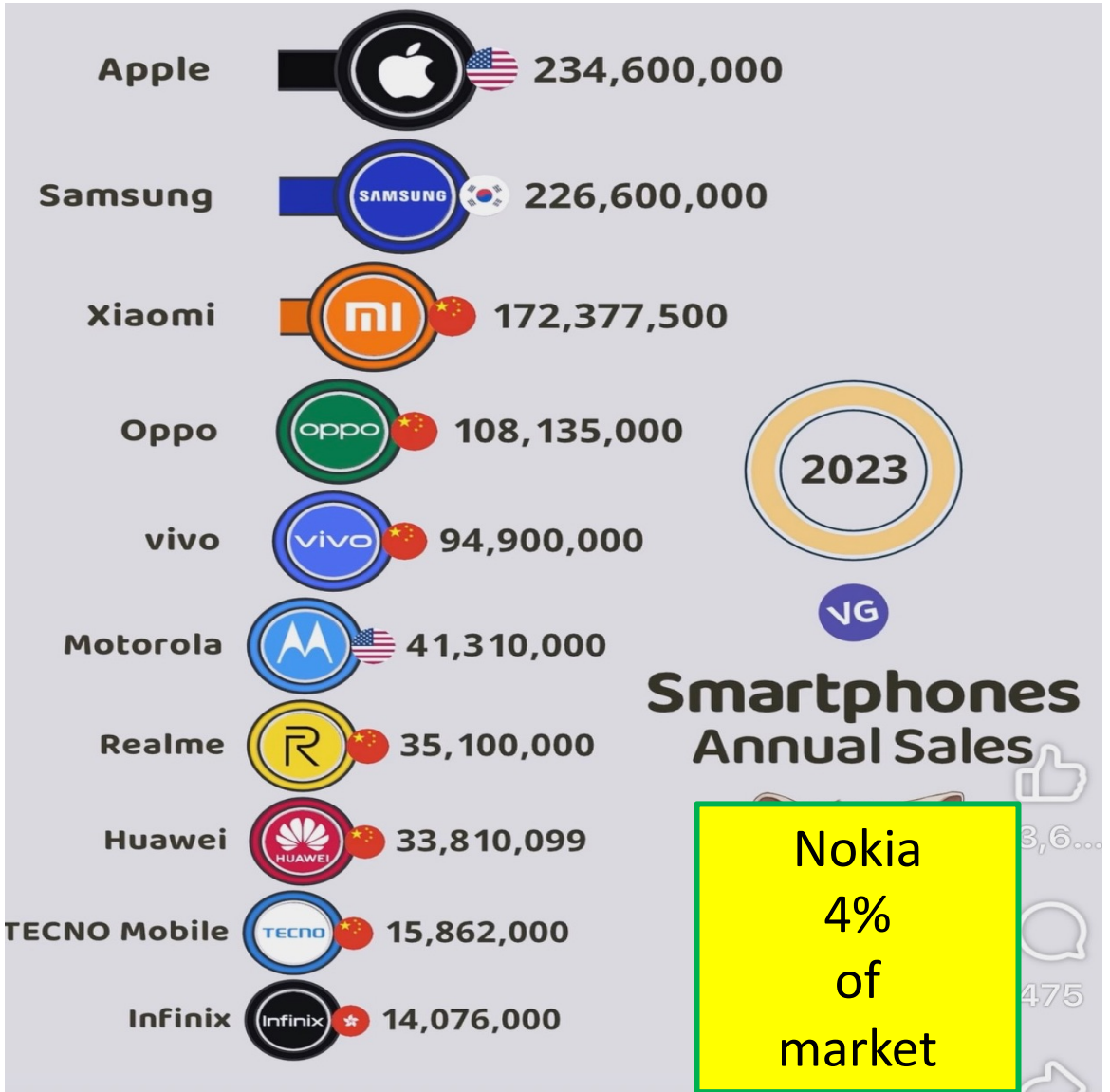
Draghi Report (2024):

- 800 billion EUR annually necessary to strengthen European innovation and industry.

European Deficits in Innovation.

- The production of mobile phones as an example for deficits in innovation in the European Union.
- Key factor: Invention of the smart phone by Apple 2007.





Source: Facebook 2024

Information und Communication Technology.

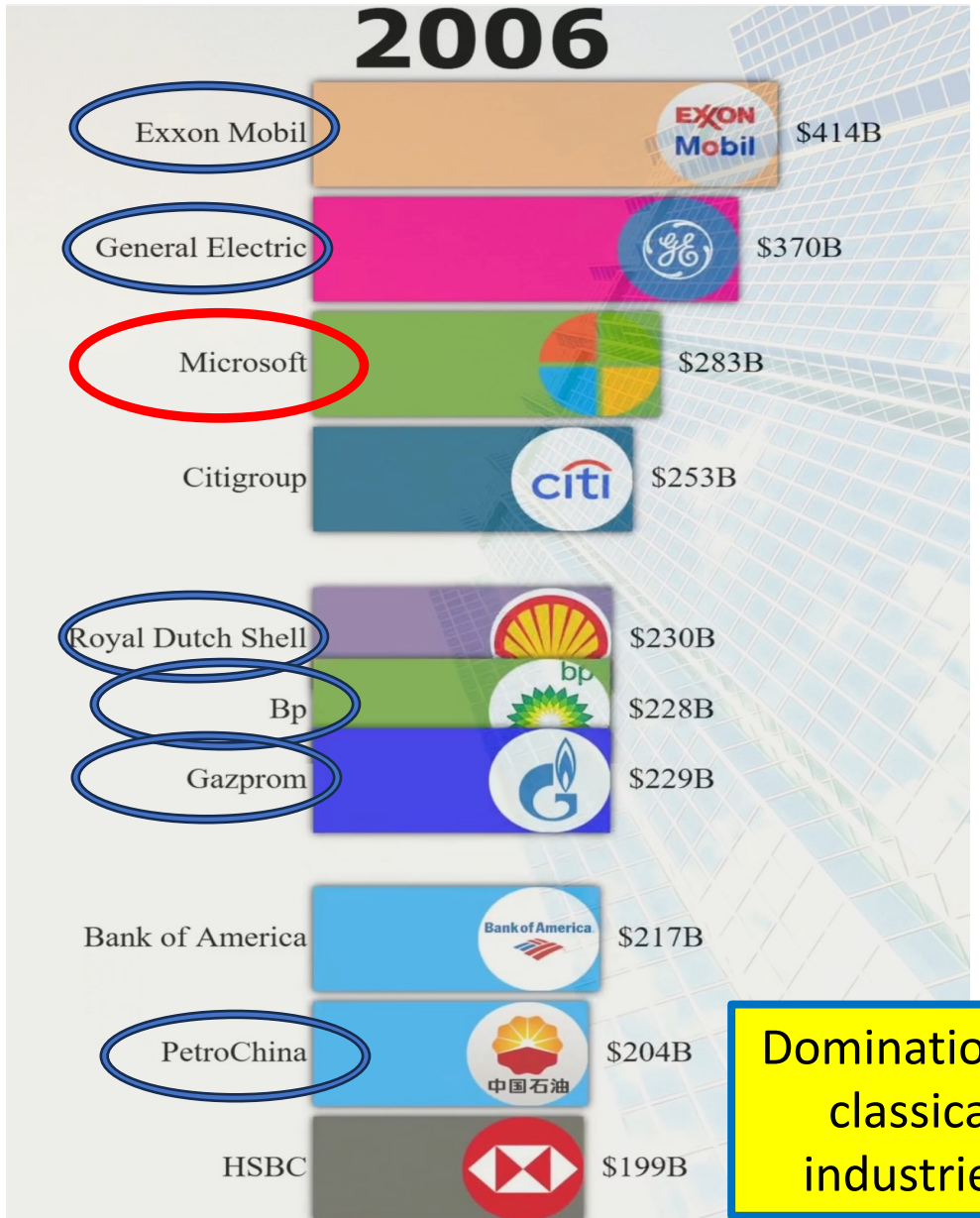


Hard Disc 5 MB, 130.000$

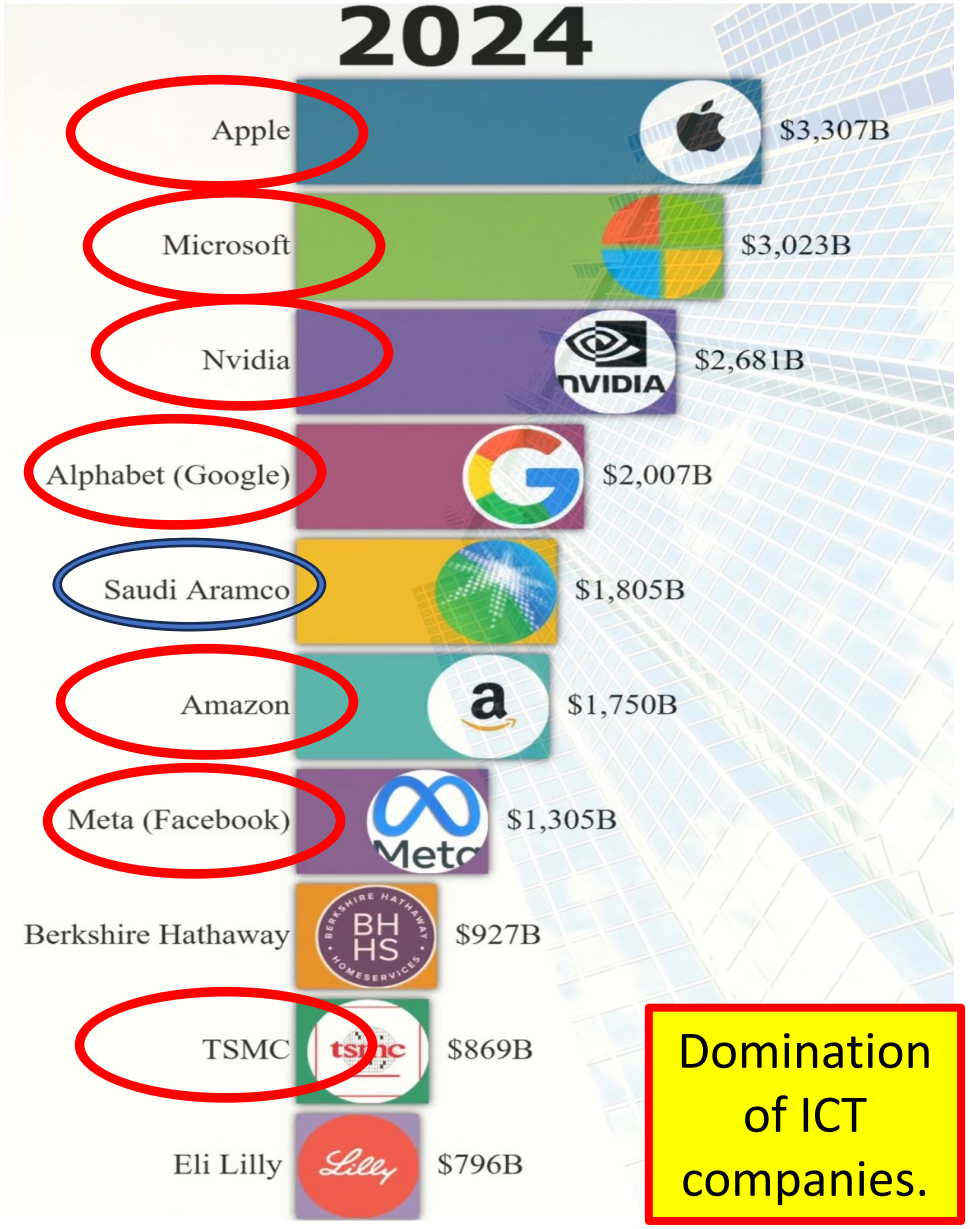


iPhone 15 storage 512.000 MB, 1.200$

The Change in Global Industrial Structures:
Largest Companies 2006/2024.



Domination of classical industries.



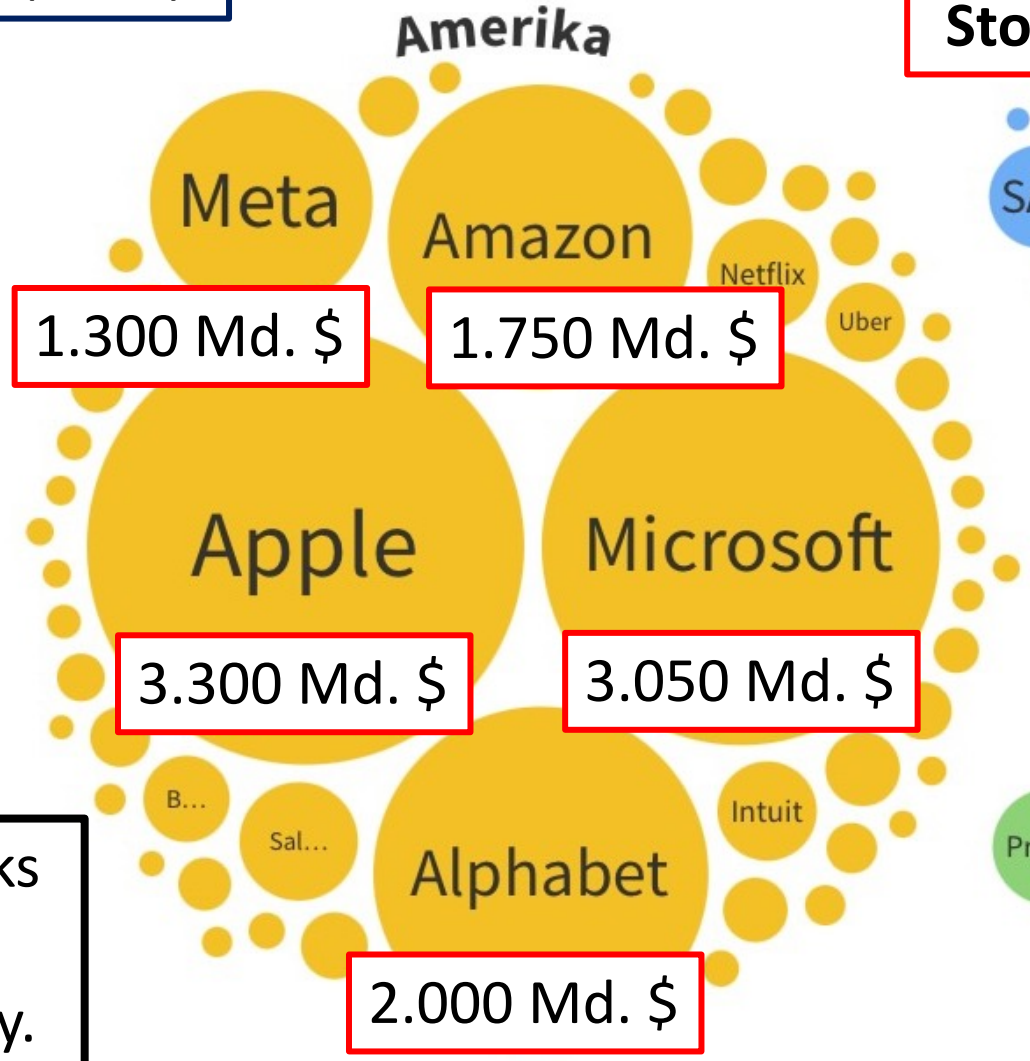
Domination of ICT companies.

Source: Facebook 2025

European Position in ICT.

Profit Meta
130 billion $ (2024).

Stock market value of HiTec platforms 2024.



1.300 Md. $

1.750 Md. $

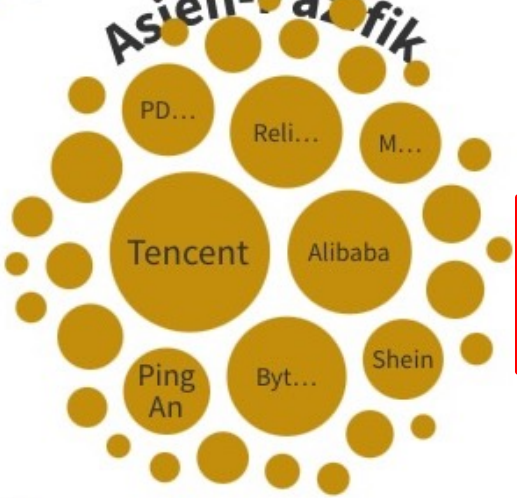
3.300 Md. $

3.050 Md. $

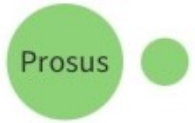
2.000 Md. $



SAP 250 billion. $



Tencent
400 billion $



Volkswagen Group :
50 billion $

Europa lacks „Digital Sovereignty.

Source: Facebook 2025

“Cognitive Revolution“ through Artificial Intelligence.

Artificial Intelligence – New Creativity?





The USA develop, the EU regulates.

Artificial Intelligence – the European Approach.

- **Artificial Intelligence:**
  - US companies spend several hundred billion $ annually on R&D in Artificial Intelligence.
  - Only 10% of the most important publications on AI are from EU (USA 60%, China 20% - rapidly rising).
  - EU primarily concerned with regulation, not innovation.
- **EU Artificial Intelligence Act (2024) foresees that professional users must provide the following documentation:**
  - Conformity assessment of AI tools, incl. CE label („Conformité Européenne“).
  - Automatically generated protocol of the actual use of AI.
  - Impact assessment on data protection.
  - Impact assessment on human rights.
  - Establishment of human control mechanisms.
  - Registration in EU data bank.
  - Guarantee of transparency of the AI system for users.
  - Risk assessment and management.

Probably impossible to handle for small enterprises.

Sascha Lobo (AI expert and journalist):
“The EU is presently destroying the future of AI in Europe.”

Source: Der Spiegel, Die Presse 2024

The EU – World Champion in Bureaucratic Perfectionism?



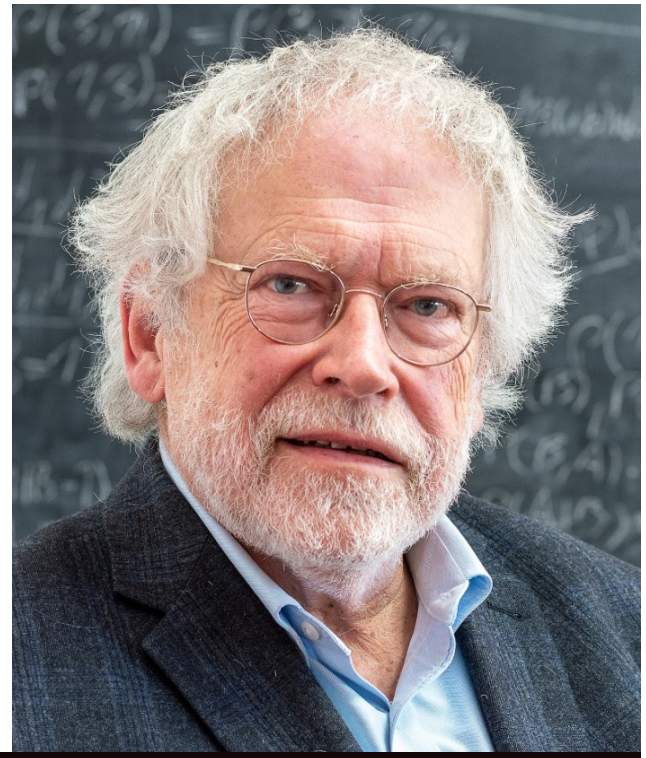
Project Omnibus 2025: Simplification of EU Regulations.

Hope for the Future.



Emmanuelle Charpentier, developed CRISP-Technology for modification of the DNA, Nobel Price for Chemistry 2020.

Anton Zeilinger, Pioneer of Quantum-Information-Science. Nobel Price for Physics 2022.





„Gene Scissor “



Quantum Entanglement for Teleportation.

Biochemical Techniques in Production of Food and Feed.

- EU plans to relax GMO restrictions (2023):
  - The plan would allow gene editing to create new varieties of crops which also in principle could be achieved through traditional breeding techniques.



Gene editing (CRISPR/Cas method).

- With this technique genes can be deleted or added from the same or similar species, accelerating and making more specific a traditional process where scientists blend different species of plant.
- It is distinct from genetic modification, which introduces DNA from foreign species.
- Crops include wheat, potatoes, tomatoes which should become more resistant against droughts and pest (like fungi).
- Would also allow to reduce the use of pesticides, fertilisers and other chemicals.

Source: European Commission, Financial Times 2023

EU proposal heavily criticized by various interest groups.